YEAR 12 MATHEMATICS EXTENSION 1

Year 11 Chapter 15
The Binomial Theorem

Lesson and Homework Pack

Revision	Counting and index laws	Year 11 · Ch 14, indices
Lesson 15A	Binomial expansions and Pascal's triangle	Past papers: 17 marks · 25 min
Lesson 15B	Further binomial expansions	Past papers: 16 marks · 25 min
Lesson 15C	The binomial theorem	Past papers: 18 marks · 25 min
Lesson 15D	Identities in Pascal's triangle	Past papers: 16 marks · 25 min
Review	Chapter 15 review exercise	Mixed practice

Each lesson runs in the same order: **concept** → **tutor demo** → **your attempt** → **worked example** to keep for homework. Then do the **textbook homework** and finish with the **past-paper set** under exam conditions. Answers and full past-paper solutions are in the separate Homework Answers booklet.

CHAPTER REVISION

Counting and index laws

A short review of what this chapter combines: factorials and ${}^nC_r$ from Chapter 14, the index laws including negative powers, and expanding brackets. Lesson 15C shows that the numbers in a binomial expansion are the ${}^nC_r$ you already know.

SUMMARY · keep this page open while you work

FACTORIALS AND ${}^nC_r$ (CHAPTER 14, 14A AND 14E)

- $n! = n(n-1)(n-2) \times \cdots \times 2 \times 1$, with $0! = 1$ and $n! = n \times (n-1)!$.
- The symbol ${}^nC_r$, or $\binom{n}{r}$, is spoken "n choose r". It is the number of r-member subsets of an n-member set, that is, the number of **unordered** selections of r objects from n objects.

$$ {}^nC_r = \binom{n}{r} = \frac{n!}{r!\,(n-r)!} = \frac{n(n-1) \times \cdots \times (n-r+1)}{1 \times 2 \times \cdots \times r} \qquad (r = 0, 1, \ldots, n) $$

- For example, ${}^8C_5 = \dfrac{8!}{3!\,5!} = \dfrac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$.
- ${}^nC_r = {}^nC_{n-r}$: choosing the r objects to take is the same as choosing the $n - r$ to leave. So ${}^nC_0 = {}^nC_n = 1$ and ${}^nC_1 = {}^nC_{n-1} = n$.
- The total number of subsets is 2^n, so ${}^nC_0 + {}^nC_1 + \cdots + {}^nC_n = 2^n$. Also ${}^nC_r$ counts words: the number of n-letter words with r letters alike of one kind and $n - r$ alike of another is $\dfrac{n!}{r!\,(n-r)!} = {}^nC_r$. For example $xxyy$ can be arranged in $\dfrac{4!}{2!\,2!} = 6$ ways.

INDEX LAWS (ADVANCED, YEAR 11), FOR $a, b \neq 0$

$$ a^m \times a^n = a^{m+n} \qquad a^m \div a^n = a^{m-n} \qquad (a^m)^n = a^{mn} \qquad (ab)^n = a^n b^n \qquad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n} $$

$$ a^0 = 1 \qquad a^{-n} = \frac{1}{a^n} \qquad a^{1/n} = \sqrt[n]{a} \qquad a^{m/n} = \left(\sqrt[n]{a}\right)^m $$

- A power applies to the number as well as to the letter: $(2x)^3 = 8x^3$, not $2x^3$.
- Take care with signs: $(-x)^n = x^n$ when n is even and $(-x)^n = -x^n$ when n is odd.
- Write a power of x in the denominator as a negative power: $\dfrac{3}{x^2} = 3x^{-2}$, and $\left(\dfrac{2}{x}\right)^3 = \dfrac{8}{x^3} = 8x^{-3}$.

EXPANDING BRACKETS (YEAR 10 AND ADVANCED)

- A **binomial** is an expression with two terms, such as $1 - x$ or $3x^4 - \frac{1}{2}x^2$.
- $(a+b)^2 = a^2 + 2ab + b^2$, $(a-b)^2 = a^2 - 2ab + b^2$ and $(a+b)(a-b) = a^2 - b^2$.
- To expand a product, multiply every term of one bracket by every term of the other, then **collect like terms**. A higher power is built one bracket at a time: $(1+x)^3 = (1+x)(1+2x+x^2) = 1 + 3x + 3x^2 + x^3$.
- The **coefficient** of a term is the number multiplying it, so the coefficient of x^2 in $5 - 3x + 7x^2$ is 7. The **constant term** (term independent of x) has no x in it.

WARM-UP · answers are in the answers booklet

1 Evaluate

a $\dfrac{7!}{5!}$

b ${}^{9}C_{3}$

c ${}^{12}C_{10}$, using ${}^{n}C_{r} = {}^{n}C_{n-r}$

d ${}^{10}C_{0} + {}^{10}C_{1} + {}^{10}C_{9} + {}^{10}C_{10}$

2 a Write ${}^{n}C_{1}$ and ${}^{n}C_{2}$ in terms of n, with no factorials.

b Find n if ${}^{n}C_{2} = 45$.

3 Simplify, writing each answer with positive indices only, or evaluate:

a $(2x^{2})^{3}$

b $\left(\dfrac{3}{x}\right)^{2} \times x^{3}$

c $\left(-\dfrac{2}{x}\right)^{3}$

d $8^{-2/3}$

4 a Expand $(2x - 3)^{2}$.

b Expand and simplify $(1 + x)(1 + 2x + x^{2})$.

c Find the coefficient of x^{2} in the expansion of $(1 + 2x)(3 - x + x^{2})$.

5 a How many different five-letter arrangements can be made from the letters $AAABB$?

b Write your answer as a value of ${}^{n}C_{r}$, and explain which choice it counts.

LESSON 15A

Binomial expansions and Pascal's triangle

A binomial expansion is the expansion of a power of a binomial, such as $(1-x)^3 = 1 - 3x + 3x^2 - x^3$. This lesson expands $(1+x)^n$ for small n, builds Pascal's triangle from its three basic properties, and uses it to expand, to approximate, to find terms in products, to solve problems about coefficients, and to expand a trinomial.

CONCEPT 1 · expansions of $(1 + x)^n$ and Pascal's triangle

BINOMIALS AND BINOMIAL EXPANSIONS

- A **binomial** is a polynomial with two terms, such as $1 - x$ or $3x^4 - \frac{1}{2}x^2$.
- A **binomial expansion** is the expansion of a power of a binomial, for example $(1-x)^3 = 1 - 3x + 3x^2 - x^3$.
- This lesson deals only with expansions of $(1+x)^n$, and the techniques that come from them. They are all based on Pascal's triangle and its three most basic properties.

SOME EXPANSIONS OF $(1+x)^n$

Each expansion is found from the one before, in two rows, with like terms in columns. To begin, $(1+x)^0 = 1$ and $(1+x)^1 = 1 + x$. Then:

$$
\begin{aligned}
(1+x)^2 &= 1(1+x) + x(1+x) \\
&= 1 + x \\
&\quad\ + x \ + x^2 \\
&= 1 + 2x + x^2
\end{aligned}
$$

$$
\begin{aligned}
(1+x)^3 &= 1(1+x)^2 + x(1+x)^2 \\
&= 1 + 2x + x^2 \\
&\quad\ + x \ + 2x^2 + x^3 \\
&= 1 + 3x + 3x^2 + x^3
\end{aligned}
$$

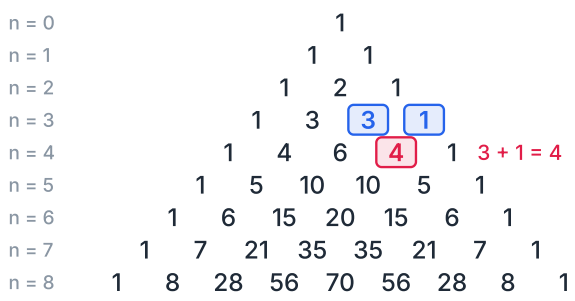
$$
\begin{aligned}
(1+x)^4 &= 1(1+x)^3 + x(1+x)^3 \\
&= 1 + 3x + 3x^2 + \boxed{1}x^3 \\
&\quad\ + x \ + 3x^2 + \boxed{3}x^3 + x^4 \\
&= 1 + 4x + 6x^2 + \boxed{4}x^3 + x^4
\end{aligned}
$$

In the last expansion, the boxed 4 is the sum of the boxed 1 and 3. Also $(1+x)^2$ has 3 terms, $(1+x)^3$ has 4 terms, and in general $(1+x)^n$ **has $n+1$ terms**, from $x^0 = 1$ to x^n. Be careful: this is inclusive counting.

PASCAL'S TRIANGLE

The coefficients of $(1+x)^n$, arranged in a table, form **Pascal's triangle**. The table (left) lines up the columns with the powers of x. The equilateral layout (right) shows the symmetry. Work from both.

n	x^0	x^1	x^2	x^3	x^4	x^5	x^6	x^7	x^8
0	1								
1	1	1							
2	1	2	1						
3	1	3	**3**	**1**					
4	1	4	6	**4**	1				
5	1	5	10	10	5	1			
6	1	6	15	20	15	6	1		
7	1	7	21	35	35	21	7	1	
8	1	8	28	56	70	56	28	8	1



1 THREE BASIC PROPERTIES OF PASCAL'S TRIANGLE

- **1.** Each row starts and ends with 1.
- **2.** Each row is reversible. That is, the rows are symmetric.
- **3.** [The addition property] Every number in the triangle, apart from the 1s, is the sum of the number directly above, and the number above and to the left.

For example, $3 + 1 = 4$. This happens because the coefficient of x^3 in $(1+x)^4$ is the sum of the coefficients of x^3 and x^2 in $(1+x)^3$: multiplying $(1+x)^3$ by 1 and by x puts both of them in the x^3 column. These properties are used in this lesson and proven formally later. The triangle can be built row by row with them.

USING THE TRIANGLE TO EXPAND

- The row for n gives the coefficients of $(1+x)^n$. To expand $(1+\text{term})^n$, replace x by the term, **in brackets**: $(1+2a)^6 = 1 + 6(2a) + 15(2a)^2 + \cdots$.
- A power applies to the number as well as to the letter: $(2a)^3 = 8a^3$, not $2a^3$.
- When the term is negative, as in $(1-x)^n$ or $\left(1-\frac{2}{3}x\right)^n$, the signs alternate.

History. Pascal did not discover the triangle, though he wrote an important treatise on it in 1653. It was known to the ancient Indian mathematician Pingala in the 2nd century BC, and later to mediaeval Persian and Chinese mathematicians. The first known occurrence in Europe is in a 1527 book on business calculations.

TUTOR DEMO **Expanding with a row of the triangle**

Use Pascal's triangle to write out the expansions of:
a $(1+3a)^4$
b $(1-x)^5$
c $\left(1-\frac{1}{2}x\right)^5$

YOUR TURN **Same skills**

Use Pascal's triangle to write out the expansions of:
a $(1+5a)^3$
b $(1-x)^3$
c $\left(1-\frac{1}{3}x\right)^4$

WORKED EXAMPLE 1 **Keep for homework**

Use Pascal's triangle to write out the expansions of:
a $(1+2a)^6$
b $(1-x)^4$
c $\left(1-\frac{2}{3}x\right)^5$

SOLUTION

a Row 6 is 1, 6, 15, 20, 15, 6, 1, so

$$\begin{aligned}(1+2a)^6 &= 1+6(2a)+15(2a)^2+20(2a)^3+15(2a)^4+6(2a)^5+(2a)^6\\ &= 1+12a+60a^2+160a^3+240a^4+192a^5+64a^6.\end{aligned}$$

b Be particularly careful with the alternating signs in this expansion.

$$\begin{aligned}(1-x)^4 &= 1+4(-x)+6(-x)^2+4(-x)^3+(-x)^4\\ &= 1-4x+6x^2-4x^3+x^4.\end{aligned}$$

c The signs also alternate in this expansion.

$$\begin{aligned}\left(1-\tfrac{2}{3}x\right)^5 &= 1+5\left(-\tfrac{2}{3}x\right)+10\left(-\tfrac{2}{3}x\right)^2+10\left(-\tfrac{2}{3}x\right)^3+5\left(-\tfrac{2}{3}x\right)^4+\left(-\tfrac{2}{3}x\right)^5\\ &= 1-\tfrac{10}{3}x+\tfrac{40}{9}x^2-\tfrac{80}{27}x^3+\tfrac{80}{81}x^4-\tfrac{32}{243}x^5.\end{aligned}$$

CONCEPT 2 · approximations

APPROXIMATING A POWER OF A NUMBER CLOSE TO 1

- Write the number as $1+(\text{a small number})$, for example $1.02^8=(1+0.02)^8$, and expand with the row of Pascal's triangle.
- The powers of a small number shrink quickly, so only the first few terms matter. Write out terms until the next ones are too small to affect the decimal place you need, then add.
- For a number just below 1, such as $0.98=1-0.02$, the signs alternate.

TUTOR DEMO **Five decimal places from a few terms**

By expanding the first few terms, find an approximation correct to five decimal places of:
a 1.01^7, using $(1+0.01)^7$
b 0.99^5, using $(1-0.01)^5$

YOUR TURN **Same method**

By expanding the first few terms, find an approximation correct to five decimal places of:
a 1.02^6, using $(1+0.02)^6$
b 0.98^4, using $(1-0.02)^4$

WORKED EXAMPLE 2 **Keep for homework**

By expanding the first few terms of $(1+0.02)^8$, find an approximation of 1.02^8 correct to five decimal places.

SOLUTION

Row 8 is 1, 8, 28, 56, 70, 56, 28, 8, 1, so

$$\begin{aligned}(1+0.02)^8 &= 1+8\times 0.02+28\times(0.02)^2+56\times(0.02)^3+70\times(0.02)^4+\cdots\\ &= 1+0.16+0.0112+0.000448+0.0000112+\cdots\\ &\doteq 1.17166.\end{aligned}$$

The remaining four terms are too small to affect the fifth decimal place.

CONCEPT 3 · terms in a product of expansions

FINDING ONE TERM OF A PRODUCT

- Expand each factor, but only as far as you need. Write powers of x in a denominator as negative powers: $\frac{5}{x} = 5x^{-1}$.
- To find the term in x^k in the product, multiply each term of the first factor by the term of the second factor that makes the power of x equal to k. Then add these products.
- The **constant term**, or **term independent of x**, is the term in x^0.

TUTOR DEMO **Constant term and term in x**

a Write out the expansion of $\left(1+\frac{3}{x}\right)^2$, then write out the first four terms in the expansion of $(1-x)^6$.

b Hence find, in the expansion of $\left(1+\frac{3}{x}\right)^2(1-x)^6$: (i) the term independent of x, (ii) the term in x.

YOUR TURN **Same method**

a Write out the expansion of $\left(1+\frac{2}{x}\right)^2$, then write out the first four terms in the expansion of $(1-x)^5$.

b Hence find, in the expansion of $\left(1+\frac{2}{x}\right)^2(1-x)^5$: (i) the term independent of x, (ii) the term in x.

WORKED EXAMPLE 3 **Keep for homework**

a Write out the expansion of $\left(1+\frac{5}{x}\right)^2$, then write out the first four terms in the expansion of $(1-x)^8$.

b Hence find, in the expansion of $\left(1+\frac{5}{x}\right)^2(1-x)^8$: (i) the term independent of x, (ii) the term in x.

SOLUTION

a $\left(1+\frac{5}{x}\right)^2 = 1+10x^{-1}+25x^{-2}$ and $(1-x)^8 = 1-8x+28x^2-56x^3+\cdots$

b Hence in the expansion of $\left(1+\frac{5}{x}\right)^2(1-x)^8$:

(i) The constant term is

$$\begin{aligned}&1\times 1+(10x^{-1})\times(-8x)+(25x^{-2})\times(28x^2)\\&=1-80+700\\&=621.\end{aligned}$$

(ii) The term in x is

$$\begin{aligned}&1\times(-8x)+(10x^{-1})\times(28x^2)+(25x^{-2})\times(-56x^3)\\&=-8x+280x-1400x\\&=-1128x.\end{aligned}$$

CONCEPT 4 · particular terms and the ratio of coefficients

WRITING DOWN ONLY THE TERMS YOU NEED

- A single term can be written straight from the row. In $(1+2kx)^6$ the term in x^3 is $20(2kx)^3 = 160k^3x^3$. You need not write out the whole expansion.
- If two coefficients are in a given ratio, write the ratio as a fraction equal to the given ratio. Cancel the common power of k and solve for k.

TUTOR DEMO **Terms and a ratio**

a Write down the terms in x^3 and x^2 in the expansion of $(1+2kx)^5$.
b Find k if these terms in x^3 and x^2 have coefficients in the ratio $3:4$.

YOUR TURN **Same method**

a Write down the terms in x^3 and x^2 in the expansion of $(1+kx)^6$.
b Find k if these terms in x^3 and x^2 have coefficients in the ratio $2:1$.

WORKED EXAMPLE 4 **Keep for homework**

a Write down the terms in x^4 and x^3 in the expansion of $(1+2kx)^6$.
b Find k if these terms in x^4 and x^3 have coefficients in the ratio $2:3$.

SOLUTION

a Row 6 is 1, 6, 15, 20, 15, 6, 1, so

$$\begin{aligned}(1+2kx)^6 &= \cdots + 15(2kx)^2 + 20(2kx)^3 + 15(2kx)^4 + \cdots \\ &= \cdots + 60k^2x^2 + 160k^3x^3 + 240k^4x^4 + \cdots\end{aligned}$$

The terms are $240k^4x^4$ and $160k^3x^3$. (Alternatively, just write down the two terms.)

b Put

$$\frac{240k^4}{160k^3} = \frac{2}{3}.$$

Then $\frac{3}{2}k = \frac{2}{3}$, so $k = \frac{4}{9}$.

CONCEPT 5 · a harder example, expanding a trinomial

TREATING PART OF A TRINOMIAL AS ONE TERM

- A trinomial such as $1+x+x^2$ can be written as $1+X$, where $X = x+x^2 = x(1+x)$.
- Expand $(1+X)^n$ with a row of Pascal's triangle. Each power $X^r = x^r(1+x)^r$ is then expanded with Pascal's triangle again.
- Finally multiply out and collect like terms. Writing like terms in columns helps.

TUTOR DEMO **A trinomial cubed**

Expand $\left(1+x+x^2\right)^3$ using Pascal's triangle, by writing $1+x+x^2 = 1+\left(x+x^2\right)$, and writing $x+x^2 = x(1+x)$.

YOUR TURN **Same method**

Expand $\left(1+x-x^2\right)^3$ using Pascal's triangle, by writing $1+x-x^2=1+\left(x-x^2\right)$, and writing $x-x^2=x(1-x)$.

WORKED EXAMPLE 5 **Keep for homework**

[A harder example] Expand $\left(1+x+x^2\right)^4$ using Pascal's triangle, by writing $1+x+x^2=1+\left(x+x^2\right)$, and writing $x+x^2=x(1+x)$.

SOLUTION

$$\begin{aligned}\left(1+x+x^2\right)^4 &= \left(1+x(1+x)\right)^4 \\ &= 1+4x(1+x)+6x^2(1+x)^2+4x^3(1+x)^3+x^4(1+x)^4 \\ &= 1+4x(1+x)+6x^2\left(1+2x+x^2\right)+4x^3\left(1+3x+3x^2+x^3\right) \\ &\quad +x^4\left(1+4x+6x^2+4x^3+x^4\right) \\ &= 1+\left(4x+4x^2\right)+\left(6x^2+12x^3+6x^4\right)+\left(4x^3+12x^4+12x^5+4x^6\right) \\ &\quad +\left(x^4+4x^5+6x^6+4x^7+x^8\right) \\ &= 1+4x+10x^2+16x^3+19x^4+16x^5+10x^6+4x^7+x^8\end{aligned}$$

TEXTBOOK HOMEWORK · Exercise 15A

FOUNDATION

1 Complete all the rows of Pascal's triangle for $n = 0,\ 1,\ 2,\ 3,\ \ldots,\ 12$. Keep this in a prominent place for use in the rest of this chapter.

2 Using Pascal's triangle to provide the binomial coefficients, give the expansions of:

a $(1+x)^6$

b $(1-x)^6$

c $(1+x)^9$

d $(1-x)^9$

e $(1+c)^5$

f $(1+2y)^4$

g $\left(1+\frac{x}{3}\right)^7$

h $(1-3z)^3$

i $\left(1-\frac{1}{x}\right)^8$

j $\left(1+\frac{2}{x}\right)^5$

k $\left(1+\frac{y}{x}\right)^5$

l $\left(1+\frac{3x}{y}\right)^4$

3 Continue the calculations of the expansions of $(1+x)^n$ at the beginning of this section, expanding $(1+x)^5$ and $(1+x)^6$ in the same manner. Keep your work in columns, so that the addition property of Pascal's triangle is clear.

4 Find the specified term in each expansion.

a For $(1+x)^{11}$:
(i) find the term in x^2,
(ii) find the term in x^8.

b For $(1-x)^7$:
(i) find the term in x^3,
(ii) find the term in x^5.

c For $(1+2x)^6$:
(i) find the term in x^4,
(ii) find the term in x^5.

d For $\left(1-\frac{3}{x}\right)^4$:
(i) find the term in x^{-1},
(ii) find the term in x^{-2}.

5 Expand $(1+x)^9$ and $(1+x)^{10}$, and show that the sum of the coefficients in the second expansion is twice the sum of the coefficients in the first expansion.

DEVELOPMENT

6 Without expanding, simplify:

a $1+3(x-1)+3(x-1)^2+(x-1)^3$

b $1-6(x+1)+15(x+1)^2-20(x+1)^3+15(x+1)^4-6(x+1)^5+(x+1)^6$

7 Find the coefficient of x^4 in the expansion of $(1-x)^4+(1-x)^5+(1-x)^6$.

8 Find integers a and b such that:

a $\left(1+\sqrt{3}\right)^5=a+b\sqrt{3}$

b $\left(1-\sqrt{5}\right)^3=a+b\sqrt{5}$

9 Verify by direct expansion, and by taking out the common factor, that:

a $(1+x)^4-(1+x)^3=x(1+x)^3$

b $(1+x)^7-(1+x)^6=x(1+x)^6$

10 Do not use a calculator in this question.

a Expand the first few terms of $(1+x)^6$, hence evaluate 1.003^6 to five decimal places.

b Similarly, expand $(1-4x)^5$, and hence evaluate 0.96^5 to five decimal places.

11 **a** (i) Expand $(1+x)^4$ as far as the term in x^2.

(ii) Hence find the coefficient of x^2 in the expansion of $(1-5x)(1+x)^4$.

b (i) Expand $(1+2x)^5$ as far as the term in x^3.

(ii) Hence find the coefficient of x^3 in the expansion of $(2-3x)(1+2x)^5$.

c (i) Expand $(1-3x)^4$ as far as the term in x^3.

(ii) Hence find the coefficient of x^3 in the expansion of $(2+x)^2(1-3x)^4$.

12 **a** When $(1+2x)^5$ is expanded in increasing powers of x, the third and fourth terms in the expansion are equal. Find the value of x.

b When $(1+x)^7$ is expanded in increasing powers of x, the sixth term is the average of the fifth and seventh terms in the expansion. Find the value of x.

13 Find the coefficient of:

a x^3 in $(3-4x)(1+x)^4$

b x in $\left(1+3x+x^2\right)(1-x)^3$

c x^4 in $\left(5-2x^3\right)(1+2x)^5$

d x^0 in $\left(1-\frac{x}{3}\right)^3\left(1+\frac{2}{x}\right)^2$

14 Determine the value of the term independent of x in the expansion of:

a $(1+2x)^4\left(1-\frac{1}{x^2}\right)^6$

b $\left(1-\frac{x}{3}\right)^5\left(1+\frac{2}{x}\right)^3$

15 a In the expansion of $(1+x)^6$:
(i) find the term in x^2,
(ii) find the term in x^3,
(iii) find the ratio of the term in x^2 to the term in x^3,
(iv) find the values of (i), (ii) and (iii) when $x=3$.

b In the expansion of $\left(1+\frac{2}{3x}\right)^7$:
(i) find the term in x^{-5},
(ii) find the term in x^{-6},
(iii) find the ratio of the term in x^{-5} to the term in x^{-6},
(iv) find the values of (i), (ii) and (iii) when $x=2$.

16 a Find the coefficients of x^4 and x^5 in the expansion of $(1+kx)^8$. Hence find k if these coefficients are in the ratio $1:4$.

b Find the coefficients of x^5 and x^6 in the expansion of $\left(1-\frac{3}{4}kx\right)^9$. Hence find k if these coefficients are equal.

17 [Patterns in Pascal's triangle]
Check the following results using the triangle you constructed in Question 1. (Some of these will be proven later.)

a The sum of the numbers in the row beginning $1,\ n,\ \ldots$ is equal to 2^n.

b If the second member of a row is a prime number, all the numbers in that row excluding the 1s are divisible by it.

c [The hockey-stick pattern]
Start at the top 1 of any column of Pascal's triangle, and go vertically down any number of rows. Then the sum of these numbers is the number in the next row down and to the right. For example, if you start at the 1 in the x^2 column, and add downwards:

$1+3+6+10+15+21=56$, which is one row down in the x^3 column.

d [The powers of 11]
If a row is made into a single number by using each element as a digit of the number, the number is a power of 11 (except that after the row $1,\ 4,\ 6,\ 4,\ 1$, the pattern gets confused by carrying).

e Find the diagonal and the column containing the triangular numbers, and show that adding adjacent pairs gives the square numbers.

ENRICHMENT

18 By writing $\left(1+x+3x^2\right)^6$ as $(1+A)^6$, where $A=x+3x^2$, expand $\left(1+x+3x^2\right)^6$ as far as the term in x^3. Hence evaluate $(1.0103)^6$ to four decimal places.

19 [The Pascal pyramid]
By considering the expansion of $(1+x+y)^n$, where $0 \leq n \leq 4$, calculate the first five layers of the Pascal pyramid.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 25 minutes **17 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q2** **1 mark**

Which of the following is the correct expansion of $(2x-1)^3$?

(A) $8x^3+12x^2+6x-1$ (B) $8x^3-12x^2+6x-1$

(C) $8x^3-4x^2+2x-1$ (D) $2x^3-12x^2+6x-1$

Hint: take out a factor so that the bracket starts with 1, then use Pascal's triangle.

PP 2 **Ascham School, 2020, Mathematics Extension 1 Trial HSC Examination, Q8** **1 mark**

The coefficient of the fourth term in the expansion of $(3x-4)^6$ is:

A. 34560 B. -34560

C. 25920 D. -25920

Hint: take out a factor so that the bracket starts with 1. How many terms does the expansion have?

PP 3 **North Sydney Boys High School, 2020, Year 12 Assessment Task 3, Mathematics Extension 1, Q2** **1 mark**

The coefficient of x^3 in the expansion of $(a+x)^6$ is 1280, what is the value of a?

A. 64 B. 4

C. 40 D. $\sqrt[3]{32}$

Hint: take out a factor so that the bracket starts with 1, and find the term in x^3 in terms of a.

PP 4 **Girraween High School, 2021, Trial Higher School Certificate Examination, Mathematics Extension 1, Q2** 1 mark

In the expansion of $(2x+k)^6$, the coefficients of x and x^2 are equal. What is the value of k?

(A) 5 (B) 6

(C) 11 (D) 12

Hint: compare Worked example 4. Take out a factor so that the bracket starts with 1.

PP 5 **Moriah College, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q9(d)** 3 marks

(i) Write down the expansion of $(1+x)^3$. 1

(ii) Hence, find the coefficient of x^4 from $(2x-1)^2(1+x)^3$ 2

Hint: in (ii), use the method of Worked example 3: which products of terms give x^4?

PP 6 **Sydney Grammar School, 2023, Trial Examination, Form VI Mathematics Extension 1, Q13(a)** 2 marks

Expand and simplify $\left(x-\frac{2}{x}\right)^4$.

Hint: take out a factor so that the bracket starts with 1.

PP 7 **Girraween High School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q16(a)** 3 marks

Find the constant term in the expression $\left(x+\frac{2}{x}\right)^6$

Hint: take out a factor so that the bracket starts with 1. The constant term is the term in x^0.

PP 8 **Abbotsleigh, 2021, Higher School Certificate Assessment Task 4, Mathematics Extension 1, Q12(c)** 2 marks

By using Pascal's triangle, or otherwise, find the coefficient of x^6 for the expansion of $(3x^2-2a)^5$.

Hint: take out a factor so that the bracket starts with 1. Which term of row 5 gives x^6?

PP 9 **Normanhurst Boys High School, 2022, Year 12 Course Assessment Task 4 (Trial Examination), Mathematics Extension 1, Q11(b)** 3 marks

Find the constant term in the expansion of $\left(3x-\frac{1}{x^2}\right)^6$.

You may leave your answers in the form $\binom{n}{k}a^b$ where $n, k, a, b \in \mathbb{Z}^+$.

Hint: take out a factor so that the bracket starts with 1, and look for the term in x^0.

LESSON 15B

Further binomial expansions

The expansion of $(x+y)^n$ uses the same coefficients as $(1+x)^n$, and because x and y are both variables, its symmetries are easier to see. This lesson expands $(x+y)^n$ with Pascal's triangle, picks out particular terms (including terms with negative powers of x), and finds terms in a product of expansions.

CONCEPT 1 · expansions of (x + y)ⁿ

A MINOR POINT ABOUT LANGUAGE

The term **polynomial** now includes terms made up of powers of any number of variables. Thus $5xy^4$ is a **monomial**, $x+y$ is a **binomial**, and $x^2+2xy+y^2$ and $x+y+z$ are **trinomials**.

SOME EXPANSIONS OF $(x+y)^n$

As before, each expansion is found from the one before, in two rows, with like terms in columns, so that the addition property is clear. To begin, $(x+y)^0=1$ and $(x+y)^1=x+y$. Then:

$$\begin{aligned}(x+y)^2 &= x(x+y)+y(x+y)\\ &= x^2+xy\\ &\quad\ +xy\ +y^2\\ &= x^2+2xy+y^2\end{aligned}$$

$$\begin{aligned}(x+y)^3 &= x(x+y)^2+y(x+y)^2\\ &= x^3+2x^2y+xy^2\\ &\quad\ +x^2y\ +2xy^2+y^3\\ &= x^3+3x^2y+3xy^2+y^3\end{aligned}$$

$$\begin{aligned}(x+y)^4 &= x(x+y)^3+y(x+y)^3\\ &= x^4+3x^3y+3x^2y^2+\boxed{1}xy^3\\ &\quad\ +x^3y\ +3x^2y^2+\boxed{3}xy^3+y^4\\ &= x^4+4x^3y+6x^2y^2+\boxed{4}xy^3+y^4\end{aligned}$$

- **First**, the coefficients are exactly the same as before, so they can be taken from Pascal's triangle.
- **Secondly**, the pattern for the indices of x and y is straightforward. The expansion of $(x+y)^3$ has four terms, and in each term the indices of x and y are whole numbers adding to 3. The expansion of $(x+y)^4$ has five terms, and in each term the indices are whole numbers adding to 4.
- More generally, in each successive expansion, beginning with $(x+y)^0=1$, the terms of the previous expansion are multiplied first by x, and then by y. So the sum of the two indices goes up by 1.

2 THE EXPANSION OF $(x+y)^n$

- The expansion of $(x+y)^n$ has $n+1$ terms, and in each term the indices of x and y are whole numbers adding to n.
- The coefficients in the expansion of $(x+y)^n$ are the same as the coefficients in the expansion of $(1+x)^n$.

The expansion of $(x+y)^n$ is called **homogeneous of degree n in x and y together**, because in each term the sum of the indices of x and y is n.

USING THE GENERAL EXPANSION

- The general expansion of $(x+y)^n$ is applied in the same way as the expansion of $(1+x)^n$. Take the coefficients from row n. The powers of the first term go down from n to 0 while the powers of the second term go up from 0 to n.
- Replace x and y by the two terms of the binomial, **in brackets**: $(2-3x)^4 = 2^4 + 4 \times 2^3 \times (-3x) + \cdots$. A power applies to the number as well as to the letter.
- When the second term is negative, the signs alternate.

TUTOR DEMO **Expanding with a row of the triangle**

Use Pascal's triangle to write out the expansions of:

a $(3-2x)^4$

b $\left(2x+\frac{1}{2}a\right)^5$

YOUR TURN **Same skills**

Use Pascal's triangle to write out the expansions of:

a $(2-5x)^3$

b $\left(3x+\frac{1}{3}a\right)^4$

WORKED EXAMPLE 6 **Keep for homework**

Use Pascal's triangle to write out the expansions of:
a $(2-3x)^4$
b $\left(5x+\frac{1}{5}a\right)^5$

SOLUTION

a Row 4 is 1, 4, 6, 4, 1, so

$$\begin{aligned}(2-3x)^4 &= 2^4+4\times 2^3\times(-3x)+6\times 2^2\times(-3x)^2+4\times 2\times(-3x)^3+(-3x)^4\\ &= 16-96x+216x^2-216x^3+81x^4.\end{aligned}$$

b Row 5 is 1, 5, 10, 10, 5, 1, so

$$\begin{aligned}\left(5x+\tfrac{1}{5}a\right)^5 &= (5x)^5+5\times(5x)^4\times\tfrac{1}{5}a+10\times(5x)^3\times\left(\tfrac{1}{5}a\right)^2\\ &\quad+10\times(5x)^2\times\left(\tfrac{1}{5}a\right)^3+5\times(5x)\times\left(\tfrac{1}{5}a\right)^4+\left(\tfrac{1}{5}a\right)^5\\ &= 3125x^5+625ax^4+50a^2x^3+2a^3x^2+\tfrac{1}{25}a^4x+\tfrac{1}{3125}a^5.\end{aligned}$$

CONCEPT 2 · particular terms, with negative powers

FINDING A PARTICULAR TERM

- Write each term of the binomial as a power of x, for example $\dfrac{1}{x^2}=x^{-2}$. Then write out the expansion with Pascal's triangle, leaving the terms unsimplified.
- Use the index laws to find the power of x in each term. From one term to the next, the power of x changes by the same amount each time.
- Pick out the term with the power you need, and simplify only that term. The **constant term**, or **term independent of x**, is the term in x^0.

TUTOR DEMO **Constant term and term in x^{-3}**

Use Pascal's triangle to write out the expansion of $\left(x+3x^{-2}\right)^6$, leaving the terms unsimplified.
Hence find:
a the term independent of x,
b the term in x^{-3}.

YOUR TURN **Same method**

Use Pascal's triangle to write out the expansion of $\left(x+2x^{-1}\right)^4$, leaving the terms unsimplified.
Hence find:
a the term independent of x,
b the term in x^{-2}.

WORKED EXAMPLE 7 **Keep for homework**

Use Pascal's triangle to write out the expansion of $\left(2x+x^{-2}\right)^6$, leaving the terms unsimplified.
Hence find:
a the term independent of x,
b the term in x^{-3}.

SOLUTION

Row 6 is 1, 6, 15, 20, 15, 6, 1, so

$$\left(2x+x^{-2}\right)^6 = (2x)^6 + 6\times(2x)^5\times\left(x^{-2}\right) + 15\times(2x)^4\times\left(x^{-2}\right)^2 + 20\times(2x)^3\times\left(x^{-2}\right)^3 + 15\times(2x)^2\times\left(x^{-2}\right)^4 + 6\times(2x)\times\left(x^{-2}\right)^5 + \left(x^{-2}\right)^6$$

a The constant term is

$$15\times(2x)^4\times\left(x^{-2}\right)^2 = 15\times 2^4\times x^4\times x^{-4} = 240.$$

b The term in x^{-3} is

$$20\times(2x)^3\times\left(x^{-2}\right)^3 = 20\times 2^3\times x^3\times x^{-6} = 160x^{-3}.$$

CONCEPT 3 · a term in a product

EXPANDING ONLY AS FAR AS YOU NEED

- To find one term in a product such as $(5+x)(2-3x)^7$, expand the power only as far as the term you need.
- Multiply each term of the first factor by the term of the expansion that gives the required power, then add these products, as in Lesson 15A.

TUTOR DEMO **Term in x² of a product**

Expand $(3-2x)^6$ as far as the term in x^2, and hence find the term in x^2 in the expansion of $(4+x)(3-2x)^6$.

YOUR TURN **Same method**

Expand $(2-x)^5$ as far as the term in x^2, and hence find the term in x^2 in the expansion of $(3+x)(2-x)^5$.

WORKED EXAMPLE 8 **Keep for homework**

Expand $(2-3x)^7$ as far as the term in x^2, and hence find the term in x^2 in the expansion of $(5+x)(2-3x)^7$.

SOLUTION

Row 7 is 1, 7, 21, 35, 35, 21, 7, 1, so

$$\begin{aligned}(2-3x)^7 &= 2^7 + 7\times 2^6\times(-3x) + 21\times 2^5\times(-3x)^2 + \cdots \\ &= 128 - 1344x + 6048x^2 - \cdots\end{aligned}$$

Hence the term in x^2 in the expansion of $(5+x)(2-3x)^7$ is

$$\begin{aligned}&5\times 6048x^2 - x\times 1344x \\ &= 28896x^2.\end{aligned}$$

TEXTBOOK HOMEWORK · Exercise 15B

FOUNDATION

1 Use Pascal's triangle to expand:

a $(x+y)^4$

b $(x-y)^4$

c $(r-s)^6$

d $(p+q)^{10}$

e $(a-b)^9$

f $(2x+y)^5$

g $(p-2q)^7$

h $(3x+2y)^4$

i $\left(a-\frac{1}{2}b\right)^3$

j $\left(\frac{1}{2}r+\frac{1}{3}s\right)^5$

k $\left(x+\frac{1}{x}\right)^6$

2 Use Pascal's triangle to expand:

a $\left(1+x^2\right)^4$

b $\left(1-3x^2\right)^3$

c $\left(x^2+2y^3\right)^6$

d $\left(x-\frac{1}{x}\right)^9$

e $\left(\sqrt{x}+\sqrt{y}\right)^7$

f $\left(\frac{2}{x}+3x^2\right)^5$

3 Simplify these expressions without expanding the brackets.

a $y^5+5y^4(x-y)+10y^3(x-y)^2+10y^2(x-y)^3+5y(x-y)^4+(x-y)^5$

b $a^4-4a^3(a-b)+6a^2(a-b)^2-4a(a-b)^3+(a-b)^4$

c $x^3+3x^2(2y-x)+3x(2y-x)^2+(2y-x)^3$

d $(x+y)^6-6(x+y)^5(x-y)+15(x+y)^4(x-y)^2-20(x+y)^3(x-y)^3$
$+15(x+y)^2(x-y)^4-6(x+y)(x-y)^5+(x-y)^6$

4 a (i) Expand $(4+x)^5$ as far as the term in x^3.
(ii) Hence find the coefficient of x^3 in the expansion of $(3-x)(4+x)^5$.
b (i) Expand $(1-2x)^6$ as far as the term in x^4.
(ii) Hence find the coefficient of x^4 in the expansion of $(1-3x)(1-2x)^6$.
c (i) Expand $(3-y)^7$ as far as the term in y^4.
(ii) Hence find the coefficient of y^4 in the expansion of $(1-y)^2(3-y)^7$.

DEVELOPMENT

5 a Expand and simplify $(x+y)^6+(x-y)^6$.
b Hence (and without a calculator) prove that

$$5^6+5^5\times 3^3+5^3\times 3^5+3^6=2^5\left(2^{12}+1\right).$$

6 Find the coefficient of:

a x^3 in $(2-5x)(x^2-3)^4$

b x^5 in $(x^2-3x+11)(4+x^3)^3$

c x^0 in $(3-2x)^2\left(x+\dfrac{2}{x}\right)^5$

d x^9 in $(x+2)^3(x-2)^7$

7 a (i) Use Pascal's triangle to expand $(x+h)^3$.

(ii) If $f(x)=x^3$, simplify $f(x+h)-f(x)$.

(iii) Hence use the definition $f'(x)=\lim\limits_{h\to 0}\dfrac{f(x+h)-f(x)}{h}$ to differentiate x^3.

b Similarly, differentiate x^5 from first principles.

8 a Show that $(3+\sqrt{5})^6+(3-\sqrt{5})^6=20608$.

b Show that $(2+\sqrt{7})^4+(2-\sqrt{7})^4$ is rational.

c If $(\sqrt{6}+\sqrt{3})^3-(\sqrt{6}-\sqrt{3})^3=a\sqrt{3}$, where a is an integer, find the value of a.

9 Show that $\dfrac{1}{(\sqrt{3}-1)^4}+\dfrac{1}{(\sqrt{3}+1)^4}=\dfrac{(\sqrt{3}+1)^4+(\sqrt{3}-1)^4}{(\sqrt{3}-1)^4(\sqrt{3}+1)^4}$ by putting the LHS over a common denominator. Then simplify the expression using Pascal's triangle.

10 Expand $(x+2y)^5$ and hence evaluate:

a $(1.02)^5$ correct to five decimal places,

b $(0.98)^5$ correct to five decimal places.

11 a Expand:

(i) $\left(x+\frac{1}{x}\right)^3$

(ii) $\left(x+\frac{1}{x}\right)^5$

(iii) $\left(x+\frac{1}{x}\right)^7$

b Hence, if $x+\frac{1}{x}=2$, evaluate:

(i) $x^3+\frac{1}{x^3}$

(ii) $x^5+\frac{1}{x^5}$

(iii) $x^7+\frac{1}{x^7}$

12 Find the coefficients of x and x^{-3} in the expansion of $\left(3x-\frac{a}{x}\right)^5$. Hence find the values of a if these coefficients are in the ratio $2:1$.

13 The coefficients of the terms in a^3 and a^{-3} in the expansion of $\left(ma + \frac{n}{a^2}\right)^6$ are equal, where m and n are non-zero real numbers. Prove that $m^2 : n^2 = 10 : 3$.

ENRICHMENT

14 a Expand $\left(x + \frac{1}{x}\right)^6$.

b If $U = x + \frac{1}{x}$, express $x^6 + \frac{1}{x^6}$ in the form $U^6 + AU^4 + BU^2 + C$. State the values of A, B and C.

15 a By starting with $\big((x + y) + z\big)^3$, expand $(x + y + z)^3$.

b Find the term independent of x in the expansion of $\left(x + 1 + x^{-1}\right)^4$.

16 [The Sierpinski triangle fractal]

a Draw an equilateral triangle of side length 1 unit on a piece of white paper. Join the midpoints of the sides of this triangle to form a smaller triangle. Colour it black. Repeat this process on all white triangles that remain. What do you notice?

b Draw up Pascal's triangle in the shape of an equilateral triangle, then colour all the even numbers black and leave the odd numbers white. What do you notice? This pattern will be more evident if you take at least the first 16 rows — perhaps use a computer program to generate 100 rows of Pascal's triangle.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 25 minutes **16 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Sydney Grammar School, 2021, Trial Examination, Form VI Mathematics Extension 1, Q2** **1 mark**

Which of the following is the coefficient of the x^2 term in the expansion of $(2x+5)^6$?

(A) 15 (B) 6000

(C) 37500 (D) 2500

Hint: use row 6 of Pascal's triangle, with $2x$ and 5 in brackets.

PP 2 **Hurlstone Agricultural High School, 2021, Higher School Certificate Assessment Task 4, Mathematics Extension 1, Q2** **1 mark**

Which of the following gives the coefficient of x^4 in the expansion of $\left(x-\sqrt{2}\right)^6$?

A. -30 B. 30

C. -60 D. 60

Hint: keep the second term $-\sqrt{2}$ in brackets, with its sign.

PP 3 **Hurlstone Agricultural High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q9** **1 mark**

What is the coefficient of x in the expansion of $\left(2x^2+\dfrac{1}{x}\right)^5$?

(A) 20 (B) 40

(C) 120 (D) 240

Hint: compare Worked example 7. Write $\dfrac{1}{x}$ as a power of x.

PP 4 **Sydney Girls High School, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(c)** 2 marks

Find the value of the constant term in the expansion of $\left(x - \frac{2}{x^2}\right)^6$.

Hint: how does the power of x change from one term to the next?

PP 5 **Knox Grammar School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(a)** 2 marks

Find the coefficient of x^4 in the expansion of $\left(x + \frac{2}{x}\right)^{10}$.

Hint: you built row 10 of Pascal's triangle in Exercise 15A Question 1.

PP 6 **Ascham School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q12(b)** 2 marks

Find the coefficient of x^2 in the expansion of $(2 + x^2)(3 - 2x)^5$.

Hint: which products of terms give x^2?

PP 7 **Pymble Ladies' College, 2023, Mathematics Extension 1 HSC Trial Examination Term 3, Q13(a)** 2 marks

What is the value of the term independent of x in the expansion of

$$\left(2-\frac{3}{x^3}\right)\left(2x+\frac{5}{x^2}\right)^6?$$

Hint: each term of the first bracket needs a partner from the expansion of the second.

PP 8 **St Aloysius' College, 2024, Trial Higher School Certificate Examination, Mathematics Extension 1, Q14(c)** 2 marks

In the expansion of the expression $(1+x)(a-bx)^{12}$, the coefficient of x^8 is zero. Find $\frac{a}{b}$ in simplest form provided neither a or b are zero.

Hint: compare Worked example 8. Which products of terms give x^8?

PP 9 **James Ruse Agricultural High School, 2023, Year 12 Trial HSC Examination, Mathematics Extension 1, Q15(b)** 3 marks

Consider the expansion of $\left(\frac{x^3}{2}+\frac{a}{x}\right)^8$. If the constant term is 5103, determine the possible value(s) of a.

Hint: which term of the expansion is the constant term?

LESSON 15C

The binomial theorem

The entries of Pascal's triangle are exactly the numbers ${}^nC_r$ from Chapter 14. This lesson proves the binomial theorem and uses the formula for ${}^nC_r$ to find a coefficient or a term without a row of the triangle, including terms with negative powers of x.

CONCEPT 1 · the binomial theorem

PASCAL'S TRIANGLE AND ${}^nC_r$

The entries in Pascal's triangle are exactly the numbers ${}^nC_r$ that arose in Chapter 14 when counting unordered selections of r elements from an n-member set. Once this has been proven below, the general binomial expansion can be written out explicitly as

$$(x+y)^n = {}^nC_0\,x^n + {}^nC_1\,x^{n-1}y + {}^nC_2\,x^{n-2}y^2 + \cdots + {}^nC_n\,y^n,$$

or, using the alternative notation $\binom{n}{r}$ for "n choose r",

$$(x+y)^n = \binom{n}{0}x^n + \binom{n}{1}x^{n-1}y + \binom{n}{2}x^{n-2}y^2 + \cdots + \binom{n}{n}y^n.$$

This result is known as the **binomial theorem**. It links the counting of Chapter 14 with the algebra of this chapter.

PROVING THE BINOMIAL THEOREM

We already know that $(x+y)^n = *\,x^n + *\,x^{n-1}y + *\,x^{n-2}y^2 + \cdots + *\,y^n$, where $*$ stands for the different coefficients. What remains is to prove that the coefficient of $x^{n-r}y^r$ is ${}^nC_r$. Take the special case $(x+y)^4$. Expanding $(x+y)(x+y)(x+y)(x+y)$ without collecting like terms, and without even using the commutative law, gives $2^4 = 16$ terms:

$$\begin{aligned}(x+y)^4 = \quad & xxxx && \text{(the term with 4 } x\text{s)}\\ & + xxxy + xxyx + xyxx + yxxx && \text{(3 } x\text{s and 1 } y\text{)}\\ & + xxyy + xyxy + yxxy + xyyx + yxyx + yyxx && \text{(2 } x\text{s and 2 } y\text{s)}\\ & + xyyy + yxyy + yyxy + yyyx && \text{(1 } x \text{ and 3 } y\text{s)}\\ & + yyyy && \text{(the term with 4 } y\text{s)}\end{aligned}$$

After collecting like terms, this becomes $x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$.

- The coefficient of x^2y^2 is 6 because there are six terms with two xs and two ys. These six terms are all the words formed with 4 letters, 2 alike of one kind and 2 of another. There are $\frac{4!}{2!\times 2!} = 6$ such words.
- More generally, when we expand $(x+y)^n$, the coefficient of x^ry^{n-r} is the number of words formed with n letters, r alike of one kind and $n-r$ alike of another. Chapter 14 showed that this number is ${}^nC_r = \frac{n!}{r!\,(n-r)!}$, the number of r-member subsets of an n-member set.

3 THE BINOMIAL THEOREM

$$(x+y)^n = {}^nC_0\,x^n + {}^nC_1\,x^{n-1}y + {}^nC_2\,x^{n-2}y^2 + \cdots + {}^nC_n\,y^n,$$

for all whole numbers n, where ${}^nC_r = \dfrac{n!}{r!\,(n-r)!}$, for $r = 0, 1, \ldots, n$.

USING THE THEOREM

- Alternatively, ${}^nC_r = \dfrac{n \times (n-1) \times \cdots \times (n-r+1)}{1 \times 2 \times \cdots \times r}$. Cancel the factorials this way instead of using a calculator.
- The notations $\binom{n}{r}$ and ${}^nC_r$ mean exactly the same thing. Do not mix the two notations in the same calculation.
- The term with y^r is ${}^nC_r\, x^{n-r} y^r$. Replace x and y by the two terms of the binomial, **in brackets**, as in Lesson 15B.

TUTOR DEMO **Coefficients from the formula**

Use the binomial theorem to calculate:
a the coefficient of x^9 in $(1+x)^{13}$,
b the term in x^4y^5 in $(x+y)^9$,
c the coefficient of A^4B^4 in $(3A-2B)^8$ (factored into primes),
d the term in $a^{n-3}b^3$ in $(a-3b)^n$.

YOUR TURN **Same skills**

Use the binomial theorem to calculate:
a the coefficient of x^6 in $(1+x)^{10}$,
b the term in x^5y^3 in $(x+y)^8$,
c the coefficient of A^3B^3 in $(2A-5B)^6$ (factored into primes),
d the term in $a^{n-2}b^2$ in $(a+4b)^n$.

WORKED EXAMPLE 9 **Keep for homework**

Use the binomial theorem to calculate:
a the coefficient of x^8 in $(1+x)^{12}$,
b the term in x^3y^4 in $(x+y)^7$,
c the coefficient of A^5B^5 in $(2A-3B)^{10}$ (factored into primes),
d the term in $a^{n-4}b^4$ in $(a-2b)^n$.

SOLUTION

a $(1+x)^{12} = \cdots + {}^{12}C_8 \times 1^4 \times x^8 + \cdots$, so the coefficient of x^8 is

$$\begin{aligned}&\frac{12!}{8! \times 4!}\\ &= \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2 \times 1}\\ &= 495.\end{aligned}$$

b $(x+y)^7 = \cdots + \binom{7}{4} \times x^3 \times y^4 + \cdots$, so the term in x^3y^4 is

$$\begin{aligned}&\frac{7!}{4! \times 3!}\, x^3y^4\\ &= \frac{7 \times 6 \times 5}{3 \times 2 \times 1}\, x^3y^4\\ &= 35x^3y^4.\end{aligned}$$

c $(2A-3B)^{10} = \cdots + {}^{10}C_5 \times (2A)^5 \times (-3B)^5 + \cdots$, so the coefficient of A^5B^5 is

$$\begin{aligned}&-\frac{10!}{5! \times 5!} \times 2^5 \times 3^5\\ &= -\frac{10 \times 9 \times 8 \times 7 \times 6}{5 \times 4 \times 3 \times 2 \times 1} \times 2^5 \times 3^5\\ &= -9 \times 4 \times 7 \times 2^5 \times 3^5\\ &= -2^7 \times 3^7 \times 7.\end{aligned}$$

d $(a-2b)^n = \cdots + \binom{n}{4} \times a^{n-4} \times (-2b)^4 + \cdots$, so the term in $a^{n-4}b^4$ is

$$\begin{aligned}&\frac{n!}{4! \times (n-4)!} \times 2^4 \times a^{n-4}b^4\\ &= \frac{n \times (n-1) \times (n-2) \times (n-3)}{4 \times 3 \times 2 \times 1} \times 2^4 \times a^{n-4}b^4\\ &= \frac{2n(n-1)(n-2)(n-3)}{3}\, a^{n-4}b^4.\end{aligned}$$

CONCEPT 2 · the particular values ${}^nC_0$, ${}^nC_1$ and ${}^nC_2$

THE FORMULAE FOR ${}^nC_0$, ${}^nC_1$ AND ${}^nC_2$

These particular formulae are important enough to be memorised:

$${}^nC_0 = \frac{n!}{0!\,n!} = 1 \qquad {}^nC_1 = \frac{n!}{1!\,(n-1)!} = n \qquad {}^nC_2 = \frac{n!}{2!\,(n-2)!} = \tfrac{1}{2}n(n-1)$$

Chapter 14 showed that ${}^nC_r = {}^nC_{n-r}$ for $r = 0, 1, 2, \ldots, n$. It follows that ${}^nC_n = 1$, ${}^nC_{n-1} = n$ and ${}^nC_{n-2} = \frac{1}{2}n(n-1)$.

4 SOME PARTICULAR VALUES OF ${}^nC_r$

- For all whole numbers n,
  $${}^nC_0 = {}^nC_n = 1, \qquad {}^nC_1 = {}^nC_{n-1} = n, \qquad {}^nC_2 = {}^nC_{n-2} = \tfrac{1}{2}n(n-1).$$
  In particular, the rows begin and end with 1.
- For all whole numbers n, and for $r = 0, 1, 2, \ldots, n$, ${}^nC_r = {}^nC_{n-r}$. That is, the rows are reversible.

TUTOR DEMO **Equations in n**

Find the value of n if:

a ${}^nC_2 = 66$

b ${}^nC_2 + {}^nC_1 + {}^nC_0 = 37$

YOUR TURN **Same method**

Find the value of n if:

a ${}^nC_2 = 28$

b ${}^nC_2 + {}^nC_1 + {}^nC_0 = 16$

WORKED EXAMPLE 10 **Keep for homework**

Find the value of n if:
a ${}^nC_2 = 55$
b ${}^nC_2 + {}^nC_1 + {}^nC_0 = 29$

SOLUTION

We know that ${}^nC_0 = 1$ and ${}^nC_1 = n$ and ${}^nC_2 = \frac{1}{2}n(n-1)$.

a

$$\frac{1}{2}n(n-1) = 55$$
$$n^2 - n - 110 = 0$$
$$(n-11)(n+10) = 0$$

Because $n \geq 0$, $n = 11$.

b

$${}^nC_2 + {}^nC_1 + {}^nC_0 = 29$$
$$\frac{1}{2}n(n-1) + n + 1 = 29$$
$$n^2 - n + 2n + 2 = 58$$
$$n^2 + n - 56 = 0$$
$$(n-7)(n+8) = 0$$

Because $n \geq 0$, $n = 7$.

CONCEPT 3 · terms with negative powers in long expansions

FINDING A TERM WHEN THE POWERS OF x ARE NEGATIVE

- Write each term of the binomial as a power of x, as in Lesson 15B. For example, $\left(x^3 - \frac{2}{x}\right)^{12} = \left(x^3 - 2x^{-1}\right)^{12}$.
- Find the power of x in the first term. From one term to the next, one factor of the first term is replaced by the second term, so the power of x changes by the same step each time. Count the number of steps r needed to reach the power you want.
- By the binomial theorem, the term after r steps is ${}^nC_r\,(\text{first term})^{n-r}(\text{second term})^r$. Simplify only that term. The **term independent of x** is the term in x^0.
- A product of two powers with the same index can sometimes be written as one power first, because $A^nB^n = (AB)^n$.

TUTOR DEMO **Constant term and term in x^8**

In the expansion of $\left(x^3 - \frac{2}{x}\right)^{12}$, find:
a the term independent of x,
b the term in x^8.

YOUR TURN **Same method**

In the expansion of $\left(x+\frac{2}{x^2}\right)^9$, find:
a the term independent of x,
b the term in x^{-6}.

WORKED EXAMPLE **Keep for homework (like Exercise 15C Question 15)**

In the expansion of $\left(x+\frac{1}{x}\right)^{20}\left(x-\frac{1}{x}\right)^{20}$, find the term independent of x. Give your answer in the form ${}^nC_r$, and also correct to four significant figures.

SOLUTION

Because $A^nB^n=(AB)^n$ and $\left(x+x^{-1}\right)\left(x-x^{-1}\right)=x^2-x^{-2}$,

$$\left(x+\frac{1}{x}\right)^{20}\left(x-\frac{1}{x}\right)^{20}=\left(x^2-x^{-2}\right)^{20}.$$

The first term $\left(x^2\right)^{20}$ has x^{40}. From one term to the next, one factor x^2 is replaced by $-x^{-2}$, so the power of x goes down by 4. The power x^0 is reached after 10 steps, so the term independent of x is

$$\begin{aligned}&{}^{20}C_{10}\times\left(x^2\right)^{10}\times\left(-x^{-2}\right)^{10}\\&={}^{20}C_{10}\times x^{20}\times x^{-20}\\&={}^{20}C_{10}\\&\approx 1.848\times 10^5.\end{aligned}$$

TEXTBOOK HOMEWORK · Exercise 15C

FOUNDATION

1 Use the result ${}^nC_r = \dfrac{n!}{r!\,(n-r)!}$ to evaluate the following. Do not use a calculator — you will need to unroll the factorial symbol. Check your answers against Pascal's triangle.

a ${}^4C_3$

b ${}^6C_3$

c ${}^9C_1$

d ${}^7C_3$

2 Repeat the previous question for these binomial coefficients. Remember the alternative notation $\binom{n}{r} = {}^nC_r$.

a $\binom{5}{2}$

b $\binom{8}{8}$

c $\binom{11}{10}$

d $\binom{10}{4}$

3 Use your calculator button $\boxed{{}^nC_r}$ to evaluate:

a ${}^{15}C_{10}$

b $\binom{13}{5}$

c ${}^{12}C_7$

d $\binom{12}{3} \div \binom{5}{2}$

e $\dfrac{{}^{15}C_8}{{}^6C_4}$

f $\dfrac{{}^{19}C_6}{{}^7C_5}$

4 a Expand $(1+x)^4$, and hence write down the values of ${}^4C_0$, ${}^4C_1$, ${}^4C_2$, ${}^4C_3$ and ${}^4C_4$.

b Hence find:

(i) ${}^4C_0 + {}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4$

(ii) ${}^4C_0 - {}^4C_1 + {}^4C_2 - {}^4C_3 + {}^4C_4$

5 Use the values of ${}^nC_r$ from Pascal's triangle to find:

a ${}^6C_0 + {}^6C_2 + {}^6C_4 + {}^6C_6$

b ${}^6C_1 + {}^6C_3 + {}^6C_5$

c ${}^2C_2 + {}^3C_2 + {}^4C_2 + {}^5C_2$

d $\left({}^5C_0\right)^2 + \left({}^5C_1\right)^2 + \left({}^5C_2\right)^2 + \left({}^5C_3\right)^2 + \left({}^5C_4\right)^2 + \left({}^5C_5\right)^2$

6 **a** Evaluate:

(i) $\binom{8}{3}$ and $\binom{8}{5}$,

(ii) $\binom{7}{4}$ and $\binom{7}{3}$.

b If $\binom{n}{3} = \binom{n}{2}$, find the value of n.

c If $\binom{12}{4} = \binom{12}{n}$, find the value of n.

7 **a** By evaluating the LHS and RHS, verify the following result for $n = 8$ and $r = 3$:

$$\binom{n}{r-1} + \binom{n}{r} = \binom{n+1}{r}$$

b Use this identity to solve each equation for n.

(i) $\binom{5}{3} + \binom{5}{4} = \binom{n}{4}$

(ii) $\binom{n}{7} + \binom{n}{8} = \binom{11}{8}$

8 Find the specified term or coefficient in each expansion.

a For $(1+x)^{18}$, find the term in x^5.

b For $(1+6x)^{12}$, find the term in x^9.

c For $(1-7x)^8$, find the term in x^3.

d For $(1+3x)^7$, find the coefficient of x^2.

9 Find the specified terms in each expansion.

a For $(2+x)^7$, find the term in x^2.

b For $\left(x+\frac{1}{2}y\right)^{14}$, find the term in x^9y^5.

c For $\left(\frac{1}{2}x-3y^2\right)^{11}$, find the term in $x^{10}y^2$.

d For $\left(a-b^{\frac{1}{2}}\right)^{20}$, find the term in a^2b^9.

10 a Find $x \neq 0$ if the terms in x^{10} and x^{11} in the expansion of $(5+2x)^{15}$ are equal.

b Find $x \neq 0$ if the terms in x^{13} and x^{14} in the expansion of $(2-3x)^{17}$ are equal.

DEVELOPMENT

11 a In the expansion of $(1+x)^{16}$, find the ratio of the term in x^{13} to the term in x^{11}.

b Find the ratio of the coefficients of x^{14} and x^{5} in the expansion of $(1+x)^{20}$.

c In the expansion of $(2+x)^{18}$, find the ratio of the coefficients of x^{10} and x^{16}.

12 a Use the binomial theorem to obtain formulae for:

(i) ${}^{n}C_{0}$

(ii) ${}^{n}C_{1}$

(iii) ${}^{n}C_{2}$

(iv) ${}^{n}C_{3}$

b Hence solve each equation for n.

(i) ${}^{9}C_{2} - {}^{n}C_{1} = {}^{6}C_{3}$

(ii) ${}^{n}C_{2} = 36$

(iii) ${}^{n}C_{2} + {}^{6}C_{2} = {}^{7}C_{2}$

(iv) ${}^{n}C_{2} + {}^{n}C_{1} = 22 - {}^{n}C_{0}$

(v) ${}^{n}C_{1} + {}^{n}C_{2} = {}^{5}C_{2}$

(vi) ${}^{n}C_{3} + {}^{n}C_{2} = 8\,{}^{n}C_{1}$

c Use the formula for ${}^{n}C_{2}$ to show that ${}^{n}C_{2} + {}^{n+1}C_{2} = n^{2}$, and verify the result on the third column of Pascal's triangle.

13 The expression $(1+ax)^n$ is expanded in increasing powers of x. Find the values of a and n if the first three terms are:

a $1+28x+364x^2+\cdots$

b $1-\frac{10}{3}x+5x^2-\cdots$

14 a In the expansion of $(2+3x)^n$, the coefficients of x^5 and x^6 are in the ratio $4:9$. Find the value of n.

b In the expansion of $(1+3x)^n$, the coefficients of x^8 and x^{10} are in the ratio $1:2$. Find the value of n.

15 In the expansion of $\left(x+\frac{1}{x}\right)^{40}\left(x-\frac{1}{x}\right)^{40}$, find the term independent of x. Give your answer in the form ${}^nC_r$, and also correct to four significant figures.

16 [Divisibility problems]

a Use the binomial theorem to show that $7^n + 2$ is divisible by 3, where n is a positive integer. (Hint: Write $7 = 6 + 1$.)

b Use the binomial theorem to show that $5^n + 3$ is divisible by 4, where n is a positive integer.

c Suppose that b, c and n are positive integers, and $a = b + c$. Use the binomial expansion of $(b + c)^n$ to show that $a^n - b^{n-1}(b + cn)$ is divisible by c^2. Hence show that $5^{42} - 2^{48}$ is divisible by 9.

17 a Use the binomial theorem to expand $(x + h)^n$.

b Hence use the definition $f'(x) = \lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$ to differentiate x^n from first principles.

ENRICHMENT

18 Use the formula ${}^nC_r = \dfrac{n!}{r!\,(n-r)!}$ to prove that if the second member of a row is a prime number, all the numbers in that row, excluding the 1s, are divisible by it.

19 [These geometrical results should be related to the numbers in Pascal's triangle.]

a Place three points on the circumference of a circle. How many line segments and triangles can be formed using these three points?

b Place four points on the circumference of a circle. How many segments, triangles and quadrilaterals can be formed using these four points?

c What happens if five points are placed on the circle?

d How many pentagons could you form if you placed seven points on the circumference of a circle?

20 [A more general form of the binomial theorem]

a Show that the binomial expansion can be written as

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots.$$

b In this form, it can be shown that the expansion is true for negative or fractional values of n, provided that the RHS is regarded as the limit of an infinite sum of powers of x. This is called the *power series expansion* of $(1+x)^n$. Assuming this, generate the binomial expansions of:

(i) $\dfrac{1}{1-x}$

(ii) $\dfrac{1}{(1-x)^2}$

(iii) $\dfrac{1}{(1+x)^2}$

(iv) $\sqrt{1+x}$

c Verify, using your expansions in parts b(i) and (ii), that $\dfrac{d}{dx}\left(\dfrac{1}{1-x}\right) = \dfrac{1}{(1-x)^2}$.

(This assumes that a power series can be differentiated term-by-term.)

PAST-PAPER HOMEWORK · mini exam

Suggested time: 25 minutes **18 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Sydney Grammar School, 2020, Trial Examination, Form VI Mathematics Extension 1, Q2** 1 mark

What is the coefficient of the x^3 term in the expansion of $(2-3x)^8$?

(A) 56 (B) -1512

(C) 1512 (D) $-48\,384$

Hint: use the binomial theorem, with $-3x$ in brackets.

PP 2 **Sydney Grammar School, 2022, Trial Examination, Form VI Mathematics Extension 1, Q7** 1 mark

What is the constant term in the binomial expansion of $\left(4x^2-\dfrac{1}{x}\right)^9$?

(A) $-344\,064$ (B) -5376

(C) 84 (D) 5376

Hint: compare the Concept 3 tutor demo. Write $\dfrac{1}{x}$ as a power of x.

PP 3 **Abbotsleigh, 2023, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q11(c)** 2 marks

Consider the expansion of $(2x-p)^9$. The coefficient of the term in x^6 is $-672\,000$.
Find the value of p.

Hint: compare Worked example 9(c).

PP 4 **North Sydney Girls High School, 2023, HSC Trial Examination, Mathematics Extension 1, Q12(b)** 3 marks

(i) Find the coefficient of x^k in the expansion of $(2+3x)^6$. 1

(ii) Hence, or otherwise, find the coefficient of x^3 in the expansion of $(2+3x)^6(1-4x^2)$. 2

Hint: in (i), compare Worked example 9(d). In (ii), which products of terms give x^3?

PP 5 **North Sydney Girls High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q11(e)** 2 marks

Find the term independent of x in the expansion of $\left(2x^2-\dfrac{3}{x}\right)^9$.

Hint: what is the power of x in the first term?

PP 6 **North Sydney Boys High School, 2020, Year 12 Assessment Task 3, Mathematics Extension 1, Q11(b)** 3 marks

Find the term independent of x in the expansion $\left(x^2-\dfrac{1}{x^2}\right)^{10}$

Hint: write $\dfrac{1}{x^2}$ as a power of x.

PP 7 **Hornsby Girls High School, 2023, Year 12 Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(d)** 3 marks

In the expansion of $(1+2x)^n$, the coefficients of x^6 and x^8 are in the ratio 7:3.
Find the value of n.

Hint: compare Exercise 15C Question 14.

PP 8 **Normanhurst Boys High School, 2021, Year 12 Course Assessment Task 4 (Trial Examination), Mathematics Extension 1, Q11(d)** 3 marks

The coefficients of the x^{17} and the x^{27} terms are equal in the expansion of

$$\left(px^3 - \frac{q}{x^2}\right)^9$$

Prove that $p - 6q = 0$, where p and q are positive integers.

Hint: how does the power of x change from one term to the next?

LESSON 15D

Identities in Pascal's triangle

Each pattern in Pascal's triangle is described by an identity on the binomial coefficients ${}^nC_r$. This lesson proves such identities in several ways: by counting subsets, by substituting into the binomial expansion, by the addition property, by differentiating and then substituting, by the formula for ${}^nC_r$, and by equating coefficients.

CONCEPT 1 · patterns and three methods of proof

IDENTITIES IN PASCAL'S TRIANGLE

There are a great number of patterns in Pascal's triangle. Some are quite straightforward to recognise and to prove, others are more complicated. They are very important in all applications of the binomial theorem, and in particular they will be important in Year 12 binomial probability.

Each pattern in Pascal's triangle is described by an identity on the binomial coefficients ${}^nC_r$. These identities sometimes have a rather forbidding appearance, so take the time to interpret each identity as some sort of pattern in Pascal's triangle. Use small values of n such as $n = 3$, $n = 4$ and $n = 5$ to work out what the identity is saying. Verify each identity in the triangle, either before or after the proof is completed.

$n\backslash r$	0	1	2	3	4	5	6	7	8	9	10
0	1										
1	1	1									
2	1	2	1								
3	1	3	3	1							
4	1	4	6	4	1						
5	1	5	10	10	5	1					
6	1	6	15	20	15	6	1				
7	1	7	21	35	35	21	7	1			
8	1	8	28	56	70	56	28	8	1		
9	1	9	36	84	126	126	84	36	9	1	
10	1	10	45	120	210	252	210	120	45	10	1

METHODS OF PROOF

Methods of proof, as well as the identities themselves, are the subject of this lesson. Each proof begins with some form of the binomial expansion. There are three basic approaches, and many identities can be proven using two or even all three of them.

- **Combinatorics.** Use the definition of ${}^nC_r$ as the number of r-member subsets of an n-member set.
- **Algebraic substitution.** Substitute values into the binomial expansion $(x+y)^n = {}^nC_0\,x^n y^0 + {}^nC_1\,x^{n-1}y^1 + {}^nC_2\,x^{n-2}y^2 + \cdots + {}^nC_n\,x^0 y^n$, or into the simpler expansion of $(1+x)^n$.
- **The addition property.** It can be proven by both methods above, and in turn it can be used to prove some important identities.

THE SUM OF A ROW

The row indexed by n has sum 2^n, that is,

$${}^nC_0 + {}^nC_1 + {}^nC_2 + \cdots + {}^nC_n = 2^n.$$

For example, the row indexed by $n = 5$ adds to $1 + 5 + 10 + 10 + 5 + 1 = 32 = 2^5$. The worked example below proves this identity by all three methods.

TUTOR DEMO **Substitution and counting**

a By substituting $x = 3$ into the binomial expansion of $(1 + x)^n$, prove that

$${}^nC_0 + 3\,{}^nC_1 + 9\,{}^nC_2 + \cdots + 3^n\,{}^nC_n = 4^n.$$

b Verify the identity in part a on the row indexed by $n = 4$.

c Use the definition of ${}^nC_r$ to explain why ${}^8C_3 = {}^8C_5$.

YOUR TURN **Same skills**

a By substituting $x = 4$ into the binomial expansion of $(1 + x)^n$, prove that

$${}^nC_0 + 4\,{}^nC_1 + 16\,{}^nC_2 + \cdots + 4^n\,{}^nC_n = 5^n.$$

b Verify the identity in part a on the row indexed by $n = 3$.

c Use the definition of ${}^nC_r$ to explain why ${}^9C_2 = {}^9C_7$.

WORKED EXAMPLE **Keep for homework (the textbook's example)**

Prove, in three different ways, that the row indexed by n has sum 2^n, that is,
$${}^nC_0 + {}^nC_1 + {}^nC_2 + \cdots + {}^nC_n = 2^n.$$

SOLUTION

Proving the result using combinatorics. Let S be an n-member set. The left-hand side ${}^nC_0 + {}^nC_1 + {}^nC_2 + \cdots + {}^nC_n$ is the sum of the numbers of 0-member subsets, 1-member subsets, 2-member subsets, ..., n-member subsets of S. That takes account of all the subsets of S, so the LHS is just the number of subsets of S.

We can count the subsets of S a different way. To form a subset, take each element in turn and choose whether it is in or out of the subset. The multiplication principle allows us to calculate the total number of ways of doing this:

1st element	2nd element	3rd element	...	nth element
2	2	2	...	2

$$\begin{aligned}\text{Hence number of subsets} &= 2 \times 2 \times 2 \times \cdots \times 2 \\ &= 2^n, \text{ which proves the result.}\end{aligned}$$

Proving the result using algebraic substitution. Start with the binomial expansion
$$(x+y)^n = {}^nC_0\, x^n y^0 + {}^nC_1\, x^{n-1}y^1 + {}^nC_2\, x^{n-2}y^2 + \cdots + {}^nC_n\, x^0 y^n.$$
Now substitute $x = y = 1$. Because $1 + 1 = 2$,
$$2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + \cdots + {}^nC_n.$$

Proving the result using the addition property. The entries in the row indexed by n are the sum of two entries in the row above, and every entry in the row above is used in exactly two sums to calculate entries in the row below. (We can regard the 1s on the ends as being $0 + 1$ and $1 + 0$.) Thus the sum of the row indexed by n is double the sum of the row above.

The first row is a single 1, which is 2^0, and the row is indexed by $n = 0$. Thus, continuing down the rows one by one, the result is clear.

CONCEPT 2 · differentiating, then substituting

A FURTHER APPROACH: DIFFERENTIATE, THEN SUBSTITUTE

- Differentiate both sides of the binomial expansion of $(1+x)^n$, and then substitute a value of x, such as $x = 1$ or $x = -1$. This is done in Questions 1, 2 and 3 of the homework, and in Question 13 of the Chapter review, which uses double differentiation. These results will be needed in Year 12 binomial probability.
- On the left, $\dfrac{d}{dx}(1+x)^n = n(1+x)^{n-1}$. On the right, the constant term ${}^nC_0$ has derivative 0, and each term ${}^nC_r\, x^r$ has derivative $r \times {}^nC_r\, x^{r-1}$.
- After substituting, each identity can be checked on a row of Pascal's triangle.

TUTOR DEMO **Differentiating a row**

Differentiate both sides of the expansion
$(1+x)^5 = {}^5C_0 + {}^5C_1\,x + {}^5C_2\,x^2 + {}^5C_3\,x^3 + {}^5C_4\,x^4 + {}^5C_5\,x^5.$
Then substitute:
a $x = 1$, to show that ${}^5C_1 + 2\,{}^5C_2 + 3\,{}^5C_3 + 4\,{}^5C_4 + 5\,{}^5C_5 = 5 \times 2^4$,
b $x = -1$, to show that ${}^5C_1 - 2\,{}^5C_2 + 3\,{}^5C_3 - 4\,{}^5C_4 + 5\,{}^5C_5 = 0$.
Check both results on the row $1\ \ 5\ \ 10\ \ 10\ \ 5\ \ 1$.

YOUR TURN **Same method**

Differentiate both sides of the expansion
$(1+x)^3 = {}^3C_0 + {}^3C_1\,x + {}^3C_2\,x^2 + {}^3C_3\,x^3.$
Then substitute:
a $x = 1$, to show that ${}^3C_1 + 2\,{}^3C_2 + 3\,{}^3C_3 = 3 \times 2^2$,
b $x = -1$, to show that ${}^3C_1 - 2\,{}^3C_2 + 3\,{}^3C_3 = 0$.
Check both results on the row $1\ \ 3\ \ 3\ \ 1$.

WORKED EXAMPLE **Keep for homework**

Differentiate both sides of the binomial expansion $(1+x)^n = {}^nC_0 + {}^nC_1\,x + {}^nC_2\,x^2 + \cdots + {}^nC_n\,x^n$, and then substitute $x = 2$, to prove that $1 \times {}^nC_1 + 2 \times 2 \times {}^nC_2 + 3 \times 2^2 \times {}^nC_3 + \cdots + n \times 2^{n-1} \times {}^nC_n = n \times 3^{n-1}$.

Then check the result on the row indexed by $n = 3$.

SOLUTION

Differentiating both sides of the expansion,

$$n(1+x)^{n-1} = {}^nC_1 + 2 \times {}^nC_2\,x + 3 \times {}^nC_3\,x^2 + \cdots + n \times {}^nC_n\,x^{n-1}.$$

Now substitute $x = 2$. Because $1 + 2 = 3$,

$$n \times 3^{n-1} = 1 \times {}^nC_1 + 2 \times 2 \times {}^nC_2 + 3 \times 2^2 \times {}^nC_3 + \cdots + n \times 2^{n-1} \times {}^nC_n.$$

Check. The row indexed by $n = 3$ is $1\ \ 3\ \ 3\ \ 1$, so

$$\text{LHS} = 1 \times 3 + 2 \times 2 \times 3 + 3 \times 4 \times 1 = 27,$$

$$\text{RHS} = 3 \times 3^2 = 27.$$

CONCEPT 3 · the formula for ${}^nC_r$, and equating coefficients

TWO MORE APPROACHES

- **Use the formula.** Use the formula ${}^nC_r = \dfrac{n!}{r!\,(n-r)!}$ to express the LHS and the RHS in terms of factorials, and then simplify. To find a common denominator, write a factorial in terms of a smaller one, for example $r! = r \times (r-1)!$ and $(n-r+1)! = (n-r+1) \times (n-r)!$. This method is used in Questions 7, 10 and 14 of the homework.
- **Equate coefficients.** Expand both sides of an algebraic identity, such as $(1+x)^n(1+x)^n = (1+x)^{2n}$, and equate the coefficients of one power of x. This method is more elaborate. A couple of examples are included in the Enrichment questions, and trial papers use it too.

TUTOR DEMO **Factorials and coefficients**

a Use the formula for ${}^nC_r$ to prove that $r \times {}^nC_r = n \times {}^{n-1}C_{r-1}$, for $1 \leq r \leq n$.
b By equating coefficients of x^2 on both sides of $(1+x)^2(1+x)^n = (1+x)^{n+2}$, prove that

$${}^nC_2 + 2\,{}^nC_1 + {}^nC_0 = {}^{n+2}C_2, \text{ for } n \geq 2.$$

Check the result in part b for $n = 3$.

YOUR TURN **Same methods**

a Use the formula for ${}^nC_r$ to prove that $(n-r) \times {}^nC_r = n \times {}^{n-1}C_r$, for $0 \leq r \leq n-1$.
b By equating coefficients of x^3 on both sides of $(1+x)^2(1+x)^n = (1+x)^{n+2}$, prove that

$${}^nC_3 + 2\,{}^nC_2 + {}^nC_1 = {}^{n+2}C_3, \text{ for } n \geq 3.$$

Check the result in part b for $n = 4$.

WORKED EXAMPLE **Keep for homework**

a Use the formula for ${}^nC_r$ to prove that $\dfrac{{}^nC_{r+1}}{{}^nC_r} = \dfrac{n-r}{r+1}$, for $0 \le r \le n-1$.

b By equating coefficients of x^2 on both sides of $(1+x)^m(1+x)^n = (1+x)^{m+n}$, prove that

$${}^mC_2 + {}^mC_1 \times {}^nC_1 + {}^nC_2 = {}^{m+n}C_2, \text{ for } m \ge 2 \text{ and } n \ge 2.$$

Check the result in part b for $m = 3$ and $n = 4$.

SOLUTION

a Using the formula, and then $(r+1)! = (r+1) \times r!$ and $(n-r)! = (n-r) \times (n-r-1)!$,

$$\begin{aligned}\frac{{}^nC_{r+1}}{{}^nC_r} &= \frac{n!}{(r+1)!\,(n-r-1)!} \times \frac{r!\,(n-r)!}{n!} \\ &= \frac{r!}{(r+1) \times r!} \times \frac{(n-r) \times (n-r-1)!}{(n-r-1)!} \\ &= \frac{n-r}{r+1}.\end{aligned}$$

b On the left, the terms in x^2 come from (term in x^k) $\times$ (term in x^{2-k}), for $k = 0, 1, 2$:

$$\begin{aligned}(1+x)^m(1+x)^n &= \left({}^mC_0 + {}^mC_1\,x + {}^mC_2\,x^2 + \cdots\right)\left({}^nC_0 + {}^nC_1\,x + {}^nC_2\,x^2 + \cdots\right) \\ &= \cdots + \left({}^mC_0 \times {}^nC_2 + {}^mC_1 \times {}^nC_1 + {}^mC_2 \times {}^nC_0\right)x^2 + \cdots\end{aligned}$$

On the right, the coefficient of x^2 in $(1+x)^{m+n}$ is ${}^{m+n}C_2$. Equating the coefficients of x^2, and using ${}^mC_0 = {}^nC_0 = 1$,

$${}^mC_2 + {}^mC_1 \times {}^nC_1 + {}^nC_2 = {}^{m+n}C_2.$$

Check. When $m = 3$ and $n = 4$, the LHS is $3 + 3 \times 4 + 6 = 21$, and the RHS is ${}^7C_2 = 21$.

TEXTBOOK HOMEWORK · Exercise 15D

FOUNDATION

1 [Substitution and differentiation]
Begin with the binomial expansion for $n = 4$,
$$(1+x)^4 = {}^4C_0 + {}^4C_1\,x + {}^4C_2\,x^2 + {}^4C_3\,x^3 + {}^4C_4\,x^4, \qquad (*)$$
and use it to prove the following results. Explain each result in terms of the row $1\ \ 4\ \ 6\ \ 4\ \ 1$ indexed by $n = 4$ in Pascal's triangle.

a (i) By substituting $x = 1$, show that ${}^4C_0 + {}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4 = 2^4$.
(ii) By substituting $x = -1$, show that ${}^4C_0 + {}^4C_2 + {}^4C_4 = {}^4C_1 + {}^4C_3$.
(iii) Hence, by using the result of part (i), show that ${}^4C_0 + {}^4C_2 + {}^4C_4 = 2^3$.

b (i) Differentiate both sides of the binomial expansion $(*)$ above.
(ii) By substituting $x = 1$, show that ${}^4C_1 + 2\,{}^4C_2 + 3\,{}^4C_3 + 4\,{}^4C_4 = 4 \times 2^3$.
(iii) By substituting $x = -1$, show that ${}^4C_1 - 2\,{}^4C_2 + 3\,{}^4C_3 - 4\,{}^4C_4 = 0$.

2 [Substitution and differentiation]
Use the general binomial expansion,
$$(1+x)^n = {}^nC_0 + {}^nC_1\,x + \cdots + {}^nC_n\,x^n \qquad (*)$$
to prove the following results. Explain each result in terms of the row $1\ \ 5\ \ 10\ \ 10\ \ 5\ \ 1$ indexed by $n = 5$ in Pascal's triangle.

a (i) By substituting $x = 1$, show that ${}^nC_0 + {}^nC_1 + \cdots + {}^nC_n = 2^n$.
(ii) By substituting $x = -1$, show that ${}^nC_0 + {}^nC_2 + {}^nC_4 + \cdots = {}^nC_1 + {}^nC_3 + {}^nC_5 + \cdots$
(iii) Hence, by using the result of part (i), show that ${}^nC_0 + {}^nC_2 + {}^nC_4 + \cdots = 2^{n-1}$.

b (i) Differentiate both sides of the binomial expansion $(*)$ above.
(ii) By substituting $x = 1$, show that $1 \times {}^nC_1 + 2 \times {}^nC_2 + \cdots + n \times {}^nC_n = n2^{n-1}$.
(iii) By substituting $x = -1$, show that $1 \times {}^nC_1 - 2 \times {}^nC_2 + \cdots + (-1)^{n-1} n \times {}^nC_n = 0$.

3 [Substitution and differentiation]
This question follows the same steps as Question 2, and uses the expansion
$(1+x)^{2n} = {}^{2n}C_0 + {}^{2n}C_1\, x + \cdots + {}^{2n}C_{2n}\, x^{2n}. \qquad (*)$

a (i) Show that ${}^{2n}C_0 + {}^{2n}C_1 + \cdots + {}^{2n}C_{2n} = 2^{2n}$.
(ii) Show that ${}^{2n}C_1 + {}^{2n}C_3 + {}^{2n}C_5 + \cdots + {}^{2n}C_{2n-1} = 2^{2n-1}$.
Check both results in Pascal's triangle, using $n = 3$ and $n = 4$.

b By differentiating both sides of the identity, show that:
(i) $1 \times {}^{2n}C_1 + 2 \times {}^{2n}C_2 + \cdots + 2n \times {}^{2n}C_{2n} = n2^{2n}$.
(ii) $1 \times {}^{2n}C_1 - 2 \times {}^{2n}C_2 + \cdots + (-1)^{2n-1}2n \times {}^{2n}C_{2n} = 0$.
Check both results in Pascal's triangle, using $n = 3$ and $n = 4$.

DEVELOPMENT

For the next three questions, you should use the definition for the symbol ${}^nC_r$ as the number of ways of choosing an r-member subset from an n-member set.

4 [Combinatorics]
Consider the set $S = \{A, B, C, D, E\}$.

a List all the subsets of $\{A, B, C, D, E\}$ containing 2 letters.

b List all the subsets of $\{A, B, C, D, E\}$ containing 3 letters.

c Consider the subset $\{A, B\}$. What letters have been omitted from the subset?

d Use the definition of ${}^nC_r$ to explain why ${}^5C_2 = {}^5C_3$.

e Explain why ${}^nC_r = {}^nC_{n-r}$ for any whole numbers n and r with $r \leq n$.

5 [Combinatorics]
Consider the set $S = \{A, B, C, D\}$.

a Write down all the subsets of S, then explain why there are exactly 2^4 of them.

b Hence explain why $\binom{4}{0} + \binom{4}{1} + \binom{4}{2} + \binom{4}{3} + \binom{4}{4} = 2^4$.

c Explain why $\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$, for any whole number n.

6 [A combinatorics proof of the addition property]
Consider the sets

$$S = \{A, B, C, D, E\} \quad \text{and} \quad U = \{A, B, C, D\}.$$

a (i) Write down all the 3-letter subsets of S that do not contain E.
(ii) Explain why this is a list of all the 3-letter subsets of U.

b (i) Write down all the 3-letter subsets of S that contain E.
(ii) Explain how to pair them up with all the 2-letter subsets of U.

c Hence explain why ${}^4C_2 + {}^4C_3 = {}^5C_3$.

d Explain why ${}^nC_{r-1} + {}^nC_r = {}^{n+1}C_r$, for any whole numbers n and r with $1 \leq r \leq n$.

7 [The addition property and the formula for ${}^nC_r$]
In this question, you will prove that if a, b, c and d are any four consecutive terms in any row of Pascal's triangle, then

$$\frac{a}{a+b}+\frac{c}{c+d}=\frac{2b}{b+c}.$$

a Consider the row $1,\ 7,\ 21,\ 35,\ 35,\ 21,\ 7,\ 1$ indexed by $n=7$. Show that the identity holds for each sequence a, b, c, d of four consecutive terms from this row.

b Choose four consecutive terms from any other row and show that the identity holds.

c Prove the identity by letting $a={}^nC_{r-1}$, $b={}^nC_r$, $c={}^nC_{r+1}$ and $d={}^nC_{r+2}$. You will need to use the addition property, then the formula ${}^nC_r=\dfrac{n!}{r!\,(n-r)!}$.

8 [Two combinatorics proofs]
We have seen in Question 2a that substituting $x=-1$ and $x=1$ into the binomial expansion proves

$${}^nC_0+{}^nC_1+{}^nC_2+{}^nC_3+\cdots=2^n. \qquad (1)$$

$${}^nC_0-{}^nC_1+{}^nC_2-{}^nC_3+\cdots=0,\ \text{for } n\geq 1. \qquad (2)$$

Here are combinatorics proofs of these results.

a Let S be an n-member set, and interpret each ${}^nC_r$ as the number of r-member subsets of S. Hence prove the first identity (1).

b To prove the second identity (2), choose a fixed element A in the set S. Pair up each subset U not containing A with the unique subset $U\cup\{A\}$ containing A.
(i) Explain why the procedure arranges all the subsets of S uniquely into pairs.
(ii) Explain why one member of each pair has an even number of members, and the other has an odd number of members.
(iii) Hence prove that ${}^nC_0+{}^nC_2+\cdots={}^nC_1+{}^nC_3+\cdots$.

9 [The hockey-stick identity]
Look at the column indexed by $r = 2$ in Pascal's triangle:

$$^2C_2 + {}^3C_2 + {}^4C_2 + {}^5C_2 + {}^6C_2 = 1 + 3 + 6 + 10 + 15 = 35 = {}^7C_3. \qquad (*)$$

The general form of this well-known *hockey-stick identity* is

$$^rC_r + {}^{r+1}C_r + {}^{r+2}C_r + \cdots + {}^nC_r = {}^{n+1}C_{r+1} \qquad (**)$$

where n and r are whole numbers with $0 \leq r \leq n$. Here are proofs of this identity using the addition property, and using a combinatorial proof.

a [A proof using the addition property]
(i) To prove the particular result $(*)$, start with the right-hand side and write

$$^7C_3 = {}^6C_3 + {}^6C_2,$$

then keep expanding the term with $r = 3$. To complete the proof, you will need to use the fact that $^3C_3 = {}^2C_2 = 1$.
(ii) Generalise this to a proof of the identity $(**)$.

b [A combinatorics proof]
(i) The number $^7C_3$ is the number of 3-member subsets of a 7-member set S. To choose a 3-member subset U of $S = \{1, 2, 3, 4, 5, 6, 7\}$, make the choice in the following way:

- First choose from S the greatest number k that will be in the subset U. This must be one of the numbers 3, 4, 5, 6 or 7, because U is to have three members, so it will have two numbers smaller than k.
- Then choose the remaining two numbers in U from the $k - 1$ possible numbers $1, 2, \ldots, k - 1$.

Explain why this method of choosing the subset yields the identity $(*)$.
(ii) Generalise this to a proof of the identity $(**)$.

10 [The formula for ${}^nC_r$]

This question involves the formula ${}^nC_r = \dfrac{n!}{r!\,(n-r)!}$.

a Prove that

$$\frac{n!}{r!\,(n-r)!} + \frac{n!}{(r-1)!\,(n-r+1)!} = \frac{(n+1)!}{r!\,(n-r+1)!}.$$

b What result have you proven?

ENRICHMENT

11 [Comparing coefficients]

By comparing coefficients of x^n on both sides of the identity $(1+x)^n(1+x)^n = (1+x)^{2n}$, show that

$$\left({}^nC_0\right)^2 + \left({}^nC_1\right)^2 + \left({}^nC_2\right)^2 + \left({}^nC_3\right)^2 + \cdots + \left({}^nC_n\right)^2 = {}^{2n}C_n.$$

Check this identity in Pascal's triangle by adding the squares of the rows indexed by $n = 1,\ 2,\ 3,\ 4,\ 5$ and 6.

12 [A combinatorial approach to Question 11]
In this question we consider binary words consisting only of the letters A and B.

a Consider a binary word consisting of $a+b$ letters, with A occurring a times and B occurring b times. Show that there are ${}^{a+b}C_a = {}^{a+b}C_b$ permutations of such a word.

b How many possible permutations are there of a binary word with $2n$ letters, if A and B both occur n times?

c A word with $2n$ letters may be split down the middle into two words of n letters. Consider the example where two As fall in the first n-letter word.
(i) How many arrangements are there of the first n-letter binary word with two As?
(ii) How many arrangements are there of the second n-letter binary word with $n-2$ of the As and 2 Bs?
(iii) How many arrangements are there of a ten-letter binary word with two As in the first half and two Bs in the second half?

13 [Comparing coefficients to prove the addition property]
Prove the addition property in Pascal's triangle by comparing coefficients of x^r on both sides of the identity
$(1+x)(1+x)^n = (1+x)^{n+1}$.

14 [An inequality proven using the formula for ${}^{n}C_{r}$]

a Prove that ${}^{2n}C_{r} < {}^{2n}C_{r+1}$, for all whole numbers n and r with $r < n$.

b Prove that ${}^{2n+1}C_{r} < {}^{2n+1}C_{r+1}$, for all whole numbers n and r with $r < n$.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 25 minutes **16 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **North Sydney Boys High School, 2020, Year 12 Assessment Task 3, Mathematics Extension 1, Q14(a)** 2 marks

Show that $\binom{n}{0}+\binom{n}{1}+\binom{n}{2}+\cdots\cdots+\binom{n}{n}=2^n$

Hint: start from the binomial expansion of $(1+x)^n$.

PP 2 **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q14(c)(i)** 2 marks

By considering $(1+x)^n = {}^nC_0 + {}^nC_1\,x + {}^nC_2\,x^2 + \cdots + {}^nC_n\,x^n$, where $n \in \mathbb{Z}^+$, show that $n2^{n-1} = {}^nC_1 + 2\,{}^nC_2 + 3\,{}^nC_3 + \cdots + n\,{}^nC_n$.

Hint: where could the number in front of each ${}^nC_r$ on the right come from?

PP 3 **Killara High School, 2021, Year 12 Mathematics Extension 1 Trial Examination, Q13(e)** 2 marks

Expand $(1+x)^{2n} + (1-x)^{2n}$ and hence, evaluate $1 + {}^{100}C_2 + {}^{100}C_4 + \cdots + {}^{100}C_{100}$

Hint: which terms cancel when the two expansions are added?

PP 4 **Blacktown Boys' High School, 2024, HSC Trial Examination, Mathematics Extension 1, Q12(d)** 2 marks

It is known that ${}^{n}C_{r} = {}^{n-1}C_{r-1} + {}^{n-1}C_{r}$ for all integers such that $1 \leq r \leq n-1$. (Do NOT prove this.)

Find ONE possible set of values for p and q such that

$${}^{2026}C_{1924} - {}^{2024}C_{100} - {}^{2024}C_{101} = {}^{p}C_{q}$$

Hint: the rows of Pascal's triangle are reversible.

PP 5 **Pymble Ladies' College, 2023, Mathematics Extension 1 HSC Trial Examination Term 3, Q14(b)** 2 marks

A Maths club has $2n$ members, half of which are male and half female.

A Pi Day committee is made up of 3 members of this club. By considering all different combinations for this committee, show that

$$n\,{}^{n}C_{2} + {}^{n}C_{3} = \frac{1}{2}\,{}^{2n}C_{3}, \quad \text{where } n \geq 3.$$

Hint: sort the possible committees by the number of males on them.

PP 6 **Kinross Wolaroi School, 2023, Year 12 Mathematics Extension 1 Trial HSC Examination, Q14(d)** 3 marks

Using the expansion of $(1+x)^3(1+x)^6$ show that:

$$\binom{3}{0}\binom{6}{6}+\binom{3}{1}\binom{6}{5}+\binom{3}{2}\binom{6}{4}+\binom{3}{3}\binom{6}{3}=\binom{9}{6}$$

Hint: write $(1+x)^3(1+x)^6$ as a single power of $(1+x)$.

PP 7 **North Sydney Boys High School, 2023, Year 12, Assessment task 3, Term 2, Mathematics Extension 1, Q14(c)** 3 marks

Given the binomial expansion of:

$$(1+x)^n = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

$$(1+x)^{n+1} = b_0 + b_1x + b_2x^2 + \cdots + b_{n+1}x^{n+1}$$

and knowing $(1+x)^{n+1} = (1+x)^n(1+x)$, find the expression, **in terms of n only**, of:

$$\frac{1}{a_0a_1a_2\cdots a_n} \times (a_0+a_1)(a_1+a_2)\cdots(a_{n-1}+a_n) \quad \text{for } n = 1, 2, 3, \ldots$$

Hint: start by writing each a_k and each b_k as a binomial coefficient.

CHAPTER 15 REVIEW

Chapter review exercise

Mixed practice from the whole chapter, Lessons 15A to 15D.

1 Write out the first five rows of Pascal's triangle.

2 Use your answer to the previous question to expand:

a $(1+x)^5$

b $(1+2x)^5$

c $(1-3x)^3$

d $(1-xy)^4$

3 a Expand $(1+7x)^5$ as far as the term in x^2.

b Hence find the coefficient of x^2 in the expansion of $(1-5x)(1+7x)^5$.

4 a Expand $(1+x)^7$.

b Hence find the first decimal place of 1.02^7.

5 Expand:

a $(3+2x)^4$

b $(5-x)^3$

c $(2x+4y)^5$

d $\left(x-\dfrac{1}{x}\right)^4$

6 Use the result ${}^nC_r = \dfrac{n!}{r!\,(n-r)!}$ to evaluate the following. Do not use a calculator — you will need to unroll the factorial symbol. Check your answers against Pascal's triangle.

a ${}^5C_3$

b ${}^7C_4$

c ${}^8C_5$

d ${}^{120}C_{60} \div {}^{119}C_{60}$

7 Use your calculator button $\boxed{{}^nC_r}$ to evaluate the following.

a ${}^{10}C_8$

b $\binom{12}{7}$

c ${}^9C_6$

d $\binom{10}{3} \div \binom{3}{2}$

e $\dfrac{{}^{10}C_5}{{}^6C_2}$

f $\dfrac{{}^{16}C_6}{{}^{10}C_5}$

8 Use your understanding of the patterns in Pascal's triangle to simplify the following expressions.

a ${}^nC_0$

b ${}^9C_4 - {}^9C_5$

c ${}^5C_0 + {}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5$

d ${}^{2n+1}C_n - {}^{2n}C_n - {}^{2n}C_{n-1}$

9 Solve ${}^nC_2 = 28$.

10 Find the coefficient of x^8 in the expansion of $(1-3x)(1+5x)^{14}$.

11 a Write out the first few terms in the expansion of $(1+x)^n$.

b By an appropriate substitution, prove that:

$$\binom{n}{0}+2\times\binom{n}{1}+4\times\binom{n}{2}+8\times\binom{n}{3}+\cdots=3^n$$

c Verify this result for the row indexed by $n=4$ in Pascal's triangle.

12 a Prove that ${}^nC_0+{}^nC_1+{}^nC_2+{}^nC_3+\cdots=2^n$.

b What is the significance of this result for Pascal's triangle?

c How may this result be interpreted as a sum of combinations?

13 **a** Differentiate the identity $(1+x)^n = {}^nC_0 + {}^nC_1\,x + \cdots + {}^nC_n\,x^n$, then make a substitution, to prove that

$$1 \times {}^nC_1 + 2 \times {}^nC_2 + \cdots + n \times {}^nC_n = n2^{n-1}.$$

b Differentiate the identity a second time, then make a substitution, to prove that

$$1 \times 2 \times {}^nC_2 + 2 \times 3 \times {}^nC_3 + 3 \times 4 \times {}^nC_4 + \cdots + (n-1) \times n \times {}^nC_n = (n-1) \times n \times 2^{n-2}.$$

c Check that the row $1,\ 5,\ 10,\ 10,\ 5,\ 1$ indexed by $n = 5$ obeys these two identities.